AVRAMOV: COMPLETE INTERSECTION DESCENDS FLAT LOCAL MAPS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written. Declared dependency: lemma-quotient-regular-ring is not carded (its statement carries an unfinished '(4) add more here.' item); recorded via external_references, not re-proved.

Allowed standard prerequisites (editorial).

- Embedding of complete local maps. more-algebra-lemma-embed-map-Noetherian-complete-local-rings (cross-chapter, named). Supplies the presentation diagram R -> S over A -> B with regular fibre.
- Three-way equivalence. lemma-perfect-map-ci (same chapter, named). The engine of both directions, applied to the embedding diagram.
- Regular quotient criterion. lemma-quotient-regular-ring (same chapter, named; NOT carded – statement carries an unfinished item). Closes the regular-sequence step for the fibre; declared as an external reference, not re-proved.

Source reference. Original reference 00MC: focused-seeds/geometry/S9-dpa-avramov/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MA: focused-seeds/geometry/S9-dpa-avramov/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MP: focused-seeds/geometry/S9-dpa-avramov/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00ON: focused-seeds/geometry/S9-dpa-avramov/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 00MG: focused-seeds/geometry/S9-dpa-avramov/reftags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00KW: focused-seeds/geometry/S9-dpa-avramov/reftags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PX: sources/stacks-dpa/pinned-original-source.tex, source lines 1836–1836. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09Q1: sources/stacks-dpa/pinned-original-source.tex, source lines 2010–2010. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07NN: focused-seeds/geometry/S9-dpa-avramov/reftags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $A \to B$ be a flat local homomorphism of Noetherian local rings. Then the following are equivalent*

(1) *$B^\wedge$ is a complete intersection,*
(2) *$A^\wedge$ and $(B/\mathfrak{m}_A B)^\wedge$ are complete intersections.*

Proof. Consider the diagram

$$\begin{array}{ccc} B & \longrightarrow & B^\wedge \\ \uparrow & & \uparrow \\ A & \longrightarrow & A^\wedge \end{array}$$

Since the horizontal maps are faithfully flat (Algebra, Lemma 00MC) we conclude that the right vertical arrow is flat (for example by Algebra, Lemma 00MP). Moreover, we have $(B/\mathfrak{m}_A B)^\wedge = B^\wedge/\mathfrak{m}_{A^\wedge} B^\wedge$ by Algebra, Lemma 00MA. Thus we may assume A and B are complete local Noetherian rings.

Assume A and B are complete local Noetherian rings. Choose a diagram

$$\begin{array}{ccc} S & \longrightarrow & B \\ \uparrow & & \uparrow \\ R & \longrightarrow & A \end{array}$$

as in More on Algebra, Lemma 07NN. Let $I = \mathrm{Ker}(R \to A)$ and $J = \mathrm{Ker}(S \to B)$. Note that since $R/I = A \to B = S/J$ is flat the map $J/IS \otimes_R R/\mathfrak{m}_R \to J/J \cap \mathfrak{m}_R S$ is an isomorphism. Hence a minimal system of generators of J/IS maps to a minimal system of generators of $\mathrm{Ker}(S/\mathfrak{m}_R S \to B/\mathfrak{m}_A B)$. Finally, $S/\mathfrak{m}_R S$ is a regular local ring.

Assume (1) holds, i.e., J is generated by a regular sequence. Since $A = R/I \to B = S/J$ is flat we see Lemma 09PX applies and we deduce that I and J/IS are generated by regular sequences. We have $\dim(B) = \dim(A) + \dim(B/\mathfrak{m}_A B)$ and $\dim(S/IS) = \dim(A) + \dim(S/\mathfrak{m}_R S)$ (Algebra, Lemma 00ON). Thus J/IS is generated by

$$\dim(S/IS) - \dim(S/J) = \dim(S/\mathfrak{m}_R S) - \dim(B/\mathfrak{m}_A B)$$

elements (Algebra, Lemma 00KW). It follows that $\mathrm{Ker}(S/\mathfrak{m}_R S \to B/\mathfrak{m}_A B)$ is generated by the same number of elements (see above). Hence $\mathrm{Ker}(S/\mathfrak{m}_R S \to B/\mathfrak{m}_A B)$ is generated by a regular sequence, see for example Lemma 09Q1. In this way we see that (2) holds.

If (2) holds, then I and $J/J \cap \mathfrak{m}_R S$ are generated by regular sequences. Lifting these generators (see above), using flatness of $R/I \to S/IS$, and using Grothendieck's lemma (Algebra, Lemma 00MG) we find that J/IS is generated by a regular sequence in S/IS. Thus Lemma 09PX tells us that J is generated by a regular sequence, whence (1) holds. □